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Teaching Note:

B&K Distributors—Calculating Return on Investment for a Web-Based Customer Portal

Note: There are free supplementary materials not included in your purchase that are necessary to teach this case. Please e-mail cases@kellogg.northwestern.edu or call (847) 491-5400 for access to these materials.

Synopsis

This teaching note illustrates the fundamentals of return on investment (ROI) analysis and sensitivity analysis for technology projects. This document is a companion to the Kellogg case entitled “B&K Distributors: Calculating Return on Investment for a Web-Based Customer Portal,” which is used in the Kellogg MBA course entitled *Managing Technology Programs and Projects* and the Kellogg executive program *Driving Strategic Results through IT Portfolio Management*.

Purpose

The purpose of the case is to teach ROI analysis best practices for technology project investments. As part of the case questions, students must decide if B&K Distributors should implement a Web-based customer portal and are assigned to assist Jim Anfield, vice president of business development for JDA Consulting, and Nancy O’Neil, B&K Distributors’s vice president of sales, in determining the feasibility of this project. They must build the final ROI projections and develop recommendations for B&K’s senior management team. Completion of this case will give the students a framework for analyzing similar investment decisions.

This case is designed to be a self-contained introduction to ROI analysis and is based on a real-life management decision for a mid-sized firm. The technology involved in the project—the implementation of a Web-based customer portal—was selected specifically because students find it easy to understand. In addition, ROI spreadsheet templates are provided so that the basic analysis follows a simple format. However, significant learnings are built into the case.

Specifically, while the structure of the templates is correct, the internal rate of return (IRR) and net present value (NPV) the students calculate assume a five-year time frame. For five years the IRR is particularly good at 40 percent, and based upon this number the vast majority of

students will recommend investing in the project. However, a Web portal infrastructure will have a useful life of approximately three years. If a three-year time frame is used, the ROI is less than the hurdle rate for the firm. In addition, the model is critically dependent on one assumption made by the management team regarding the market penetration. Sensitivity analysis reveals the problems with the assumptions and indicates how the project might be modified to improve the expected outcome.

The key learning of this case is that ROI analysis for technology projects is not a straightforward activity—several factors need to be analyzed in order to conduct a thorough review of the feasibility of the investment. For example, financial elements such as payback period, sensitivity analysis, and outcome ranges are essential to the decision-making process. The case discussion emphasizes the importance of assumptions and the range of possible outcomes.

Case Questions

The case assignment is as follows:

Your team has been assigned to help Anfield and O’Neil build the final ROI projection and develop a presentation and recommendation for B&K’s senior management team. Time is short, and to help your team, Anfield has provided a copy of the ROI template his consulting firm uses. This template is included in Case Exhibits 14A–14D, and is also available in the Excel file accompanying this case. When performing your analysis, assume a tax rate of 38 percent and a cost of capital of 12 percent.

Additional related questions are:

1. *What is the return on investment for the proposed Web portal project?*
2. *How much value is generated from cost savings associated with the project?*
3. *How much value lies in the improvement of revenues?*
4. *What additional factors should be included in the ROI analysis?*
5. *If you were B&K’s CEO, would you move forward with the project? Why or why not?*

An additional disclaimer is given to students, who may focus only on the answers to these questions: *These questions are intended to help your analysis. However, your final solution to the case and recommendations to management should not necessarily be limited to the answers to these questions or the assumptions in the case.*

Analysis

FRAMEWORK FOR SOLVING THE CASE

Students will work through the case in groups. They will likely approach the case by using the provided templates and analyzing the resulting IRR,¹ NPV, and payback period. Students will then try to qualitatively interpret the resulting figures.

DETAILED SOLUTION

Step 1: *Understanding the Case Exhibits*

The following case exhibits provide data to populate spreadsheet templates found in Case Exhibits 14A through 14D.

Case Exhibit 1: A, B, and C Tier Order Data (Current State) shows a table representation of the percentage of accounts and the number of orders received for each customer tier level at present (without the implementation of the Web-based customer portal). The students are expected to interpret and study the information and be able to update the ROI spreadsheet templates accordingly. This exhibit provides specific information for Case Exhibit 14C and is the starting point for Case Exhibit 14B.

Case Exhibit 2: Revenue and Cost Data Per Order (Current State) shows a table representation of the average order size across all three customer tiers and the associated percentage of cost of goods sold (COGS) at present (without the implementation of the Web-based customer portal). The students are expected to interpret and study the information and be able to update the ROI spreadsheet templates accordingly. This exhibit provides specific information for Case Exhibit 14C and is the starting point for Case Exhibit 14B.

Case Exhibit 3: Fixed Transaction Cost Per Order by Channel (Current State) shows a table representation of the transaction cost of an order based on the channel used and the percentage of occurrence at present (without the implementation of the Web-based customer portal). The students are expected to interpret and study the information and be able to update the ROI spreadsheet Case Exhibit 14D accordingly.

Case Exhibit 4: Web Portal Prototype shows a screen prototype of the expected Web-based customer portal including all expected features and functionalities. The students may review this prototype to help them understand the details and determine the feasibility of the project.

Case Exhibit 5: Franchise Penetration Over Time shows a graphical representation of the expected market penetration rate over a five-year period should B&K Distributors continue with the implementation of the Web-based customer portal. The students are expected to interpret and study the information and be able to update the ROI spreadsheet Case Exhibit 14B accordingly.

Case Exhibit 6: Projected Tier Migration Over Time shows a graphical representation of the projected migration of customers from lower-tier levels to higher-tier levels over a five-year period should B&K Distributors continue with the implementation of the Web-based customer

¹ Note that the term ROI is used loosely in industry. In this teaching note, we focus on the best practice of calculating IRR. There is a standard and unambiguous definition of IRR, and when we use the term ROI it can be assumed to be interchangeable with IRR throughout.

portal. The students are expected to interpret and study the information and be able to update the ROI spreadsheet Case Exhibit 14B accordingly.

Case Exhibit 7: Order Size Improvement Over Time shows a graphical representation of the projected improvement in customer order size over a five-year period should B&K Distributors continue with the implementation of the Web-based customer portal. The students are expected to interpret and study the information and be able to update the ROI spreadsheet Case Exhibit 14B accordingly.

Case Exhibit 8: Cost Per Order by Channel shows a table representation of costs per order by channel (telephone, fax/mail, and Web). The students are expected to interpret the information and be able to update the ROI spreadsheet Case Exhibit 14D accordingly.

Case Exhibit 9: Order Channel Migration Over Time shows a graphical representation of the projected migration of customers from phone and fax to the Web channel over a five-year period should B&K Distributors continue with the implementation of the Web-based customer portal. The students are expected to interpret the information and be able to update the ROI spreadsheet Case Exhibit 14B accordingly.

Case Exhibit 10: Implementation Costs shows a table representation of the projected costs should B&K Distributors continue with the implementation of the Web-based customer portal. The students are expected to interpret the information and be able to update the ROI spreadsheet Case Exhibit 14A accordingly.

Case Exhibit 11: Project Phases shows a graphical representation of the expected development and operational stages of the Web-based customer portal project. The students may review the information to help them understand the details and determine the feasibility of the project.

Case Exhibit 12: Web Host Infrastructure shows a graphical representation of the expected technical infrastructure associated with the Web-based customer portal project. The students may review the information to help them understand the details and determine the feasibility of the project.

Case Exhibit 13: Annual Support Costs shows a graphical representation of the projected support costs per year should B&K Distributors continue with the implementation of the Web-based customer portal. The students are expected to interpret the information and be able to update the ROI spreadsheet Case Exhibit 14A accordingly.

Case Exhibits 14A–14D: ROI Spreadsheet Templates include a summary page, upside projection, base case projection, and average transaction cost matrix. (See below for further description.)

Step 2: Updating the ROI Spreadsheet Templates

The students are expected to update the ROI spreadsheet templates based on the previously discussed exhibits. If entered correctly, the students' resulting figures should be identical to those presented in this teaching note and the accompanying spreadsheet solution file. Students work back-to-front completing Exhibits 14D, 14C, 14B, and then finally 14A.

Case Exhibit 14D: Average Transaction Cost Matrix is the ROI spreadsheet template for entering the expected transaction costs associated with each order channel and the projected percentage of orders that will be received through these channels. Students fill in the blue-shaded region of the template based on previously provided information in Case Exhibit 9: Order Channel Migration Over Time to come up with the proper results.

Case Exhibit 14C: Base Case Projection is the ROI spreadsheet template for entering the expected growth, penetration, tier structure, and order size for B&K Distributors in the next five years without the implementation of the Web-based customer portal. Students fill in the blue-shaded region of the template based on previously provided information in the case write-up and exhibits to come up with the proper revenue results.

Case Exhibit 14B: Upside Projection is the ROI spreadsheet template for entering the expected growth, penetration, tier structure, and order size for B&K Distributors in the next five years with the implementation of the Web-based customer portal. Students fill in the blue-shaded region of the template based on previously provided information in the case write-up and exhibits to come up with the proper revenue results.

Note that Case Exhibits 14B and 14C are very similar. These ROI spreadsheet templates are meant to help the students assess and compare the projected revenue streams for the next five years with or without the implementation of the Web-based customer portal.

Case Exhibit 14A: Summary Page is the ROI spreadsheet template for entering the expected incremental revenue and costs associated with the implementation of the Web-based customer portal. Once Case Exhibits 14B and 14C are completed properly, data from these two templates will feed into this spreadsheet, displaying the projected incremental revenue of the project. In addition, students fill in the blue-shaded region of the template based on previously provided information in the case write-up and Case Exhibits 10 and 13 to come up with the proper cost results. For this case, the discount rate is 12 percent, the tax rate is 38 percent, and the support cost inflation rate is 3 percent. The summary page will then display the total incremental cash flow for the five-year period, the expected payback period, the IRR, and the NPV for the project.

Exhibit TN1 shows the summary five-year IRR analysis. The project IRR is 40.6% and the NPV is positive \$346,000.

Students often make one of two errors when they first complete Exhibit 14A. First, they may not consistently put all of the units in thousands of dollars. Second, they may not put the costs of the project in as negative numbers. If this is the mistake, the IRR will not calculate.

At this point, the solution to the case may look very simple, and some students may be tempted to establish their final recommendations based on the results of Case Exhibit 14A. However, astute students should realize that a straightforward answer is not sufficient for this case. The students are encouraged to use their judgment and analyze the feasibility of the projected information given in the case exhibits. One goal of this case is for the students to submit realistic ranges for the initial projections—creating best-case, expected-case, and worst-case scenarios.

Step 3: Conducting the Sensitivity Analysis

To be able to determine the best-case, expected-case, and worst-case scenarios, students should conduct sensitivity analyses on the information provided in the case. There are four main variables for increasing the revenue of B&K Distributors:

1. Market penetration
2. Tier migration
3. Order size improvement
4. Order channel migration

Anfield and O'Neil are making an assumption that by implementing the Web-based customer portal, the company will see dramatic improvements in these four areas. In a span of five years, they project market penetration to increase from 50 percent to 70 percent, tier migration to move to a 33 percent A/60 percent B/7 percent C mix, order size to increase from 3 percent to 6 percent, and channel migration to move to a mix of 20 percent Web/32 percent fax or mail/48 percent phone. However, the main question is: how realistic are these projections?

If the students plug the projected figures into Case Exhibit 14B: Upside Projection, it is easy to determine the incremental revenues that can be achieved by implementing the Web-based customer portal. However, once any of these four main revenue-generating targets are not met fully, the resulting revenue streams may change significantly. Several external factors may affect the outcome of the project. Within the five-year period the economy may change: competitors may enter the market and steal market share, new ordering channels may emerge, customer preferences and needs may change, etc. There are myriad scenarios that may occur that could dramatically change the results of the project. By conducting sensitivity analysis, the students will be able to address the following questions:

1. Which revenue assumption is the most critical for B&K Distributors?
2. How flexible are the projections to the different scenarios that may or may not occur?

In addition, there are two main components for managing the costs associated with the project, implementation costs and annual support costs.

Similar to the revenue assumption targets resulting from the Web-based customer portal, several external factors may also affect the cost outcome of the project. For example, implementation may take longer than planned, or maintenance costs may increase in the future. The sensitivity analysis will also help the students to determine the flexibility of the projections in handling over- and under-budget scenarios.

To illustrate how sensitivity analysis can be applied to this case, **Exhibit TN2** and **Exhibit TN3** are five-year single-variable sensitivity and five-year multiple-variable sensitivity spreadsheets that may be presented in class.

The spreadsheets in Exhibit TN2 and Exhibit TN3 were created using the Table function of Microsoft Excel. The **Appendix** at the end of this teaching note provides detailed instructions for sensitivity analysis using the data function in Microsoft Excel, and these instructions can be handed out at the end of the case debrief so that students can use the technique in the future.

After reviewing the different possible scenarios for the project, it is important to select the option(s) to present to senior management. Since there is a large range of possible end results, the students must learn how to judge (based on their comprehension of the different factors surrounding the project) which they believe to be the most realistic scenario. The option(s) presented must then be supported by sufficient research and analysis.

Once the students learn and understand how to conduct a sensitivity analysis for the case, they will realize that the resulting IRR and NPV of the initiative depend largely on meeting the revenue and cost targets proposed at the start of the project. By understanding this, the students can then revisit their original assumptions and determine if these are feasible given current technological trends, economic conditions, and other factors that may impact the project. They may decide to update their projections if they feel that these are too aggressive or too conservative. At the same time, they may also provide qualitative solutions such as conducting market research and implementing strategic action plans to reach these goals.

Step 4: Assessing the Appropriate Time Frame

At this point, it is also essential for the students to realize that time is a crucial element to this project. One critical aspect of this case is that the ROI spreadsheet templates are presented with a five-year time frame. This may lead students to accept the duration without considering if this time frame is appropriate for the project at hand. The students should learn to question this basic assumption: Will technology or economic conditions change so drastically that a five-year plan for such an initiative is not feasible? Will two or three years be more feasible time frames?

It is important to realize that changing the time frame assumption for the case will drastically impact the resulting IRR and NPV of the project. To illustrate how time frame analysis can be applied to this case, included with this teaching note are the three-year single-variable sensitivity (**Exhibit TN4**) and three-year multiple-variable sensitivity (**Exhibit TN5**) spreadsheets that may be presented in class. Note that for both cases the IRR is never above the hurdle rate.

Case Question Solutions

The assignment can be completed by answering the following questions:

1. What is the return on investment for the proposed Web portal project?

Given a discount rate of 12 percent, tax rate of 38 percent, and support cost inflation rate of 3 percent, the five-year IRR for the project is 40.6 percent with an NPV of \$346,000. For the detailed figures, please refer to the solutions spreadsheet.

2. How much value is generated from cost savings associated with the project?

The expected transaction cost savings per year (without factoring in the investment costs of the project) are as follows:

	Year 1	Year 2	Year 3	Year 4	Year 5
Savings	\$ 0.25	\$ 0.79	\$ 1.31	\$ 0.73	\$ 1.51
Transactions	17,520	20,004	21,787	23,108	23,571
<i>Total</i>	<i>\$ 4,397.52</i>	<i>\$ 15,770.60</i>	<i>\$ 28,506.48</i>	<i>\$ 16,757.50</i>	<i>\$ 35,604.60</i>

These figures are based on the forecasted order channel migration rate (20 percent Web/32 percent fax or mail/48 percent phone) and include the appropriate annual inflation cost increases (1 percent Web/3 percent fax or mail/3 percent phone). The NPV of the cost savings for the five-year period is \$67,641.64.

3. How much value lies in the improvement of revenues?

The expected incremental revenues per year (without factoring in the investment costs of the project) are as follows (in thousands of dollars):

	Year 1	Year 2	Year 3	Year 4	Year 5
Upside	\$ 4,643	\$ 5,619	\$ 6,487	\$ 7,293	\$ 7,886
Baseline	3,608	3,828	4,061	4,308	4,569
Incremental	1,035	1,791	2,426	2,986	3,316

These figures are based on the forecasted market penetration (70 percent), tier migration (33 percent A/60 percent B/7 percent C), and order size improvement (6 percent). The present value (PV) of the incremental revenues for the five-year period is \$7,858,000.

4. What additional factors should be included in the ROI analysis?

Several factors should be included in the ROI analysis of this project. First, the revenue and cost assumptions have to be reviewed for feasibility and presented to senior management for approval. Second, a sensitivity analysis has to be conducted to see how best-case, middle-case, and worst-case scenarios affect the final IRR and NPV results. In addition, the team should also identify the appropriate time frame for the ROI analysis. Third, based on the ranges of results possible, the team must identify the appropriate scenario to present to senior management. This final proposal will be used as the implementation plan for the project and the measuring standard for the end results.

5. If you were B&K's CEO, would you move forward with the project? Why or why not?

The Web-based customer portal is a high-risk project and should either be postponed, modified, or implemented cautiously with the full support of senior management. By conducting a sensitivity analysis, it can be observed that the incremental value of the project relies heavily on meeting the target assumptions of the project, particularly the expected market penetration rate of 70 percent. If this expected target is not met, the project will not meet the hurdle rate of the firm. Furthermore, by changing the time frame from five years to three years, the project appears even less attractive even if all revenue targets are met.

Teaching the Case

COMMON STUDENT APPROACHES TO THE SOLUTION

All of the student groups should be able to fill out the ROI template case exhibits and calculate the project ROI quite easily. Based on the results of the ROI spreadsheet templates, the majority of students will most likely recommend the implementation of the Web-based customer portal because the project has an ROI of 40.6 percent. This is partly due to the unique setup of the ROI spreadsheet templates—these are initially presented with a five-year duration, so some students may fall into the trap of accepting this time frame without question. The solution based upon a more realistic time frame of three years is discussed below. Most astute students will

probably be able to implement a basic form of sensitivity analysis. A good analysis is one that takes into consideration the time frame of the analysis, the assumptions underlying the model, and the range of possible outcomes.

TEACHING APPROACH

During the class prior to teaching this case, the instructor may select one of the student groups to prepare a twenty-minute presentation of their solution to the case. The student team can present the case at the beginning of the class followed by a question-and-answer session. This should get the rest of the class thinking about their solutions and how or why they differ from the one presented. After the presentation, the instructor may ask the remaining groups for their solutions and list them on the blackboard.

Next, turn the class discussion over to the case questions and walk through the step-by-step solutions in the ROI spreadsheet templates as detailed in the Analysis portion of this teaching note. The initial payback period, IRR, and NPV calculations are fairly straightforward, and the results the students get should be more or less consistent with the solutions provided in this note. Verify that the entire class understands the process and solutions before proceeding any further.

After making sure that the class completely understands the initial ROI calculations, present how sensitivity analysis can be used in this case to come up with an improved and more comprehensive analysis of the case. Begin the discussion by showing the solution spreadsheet of Case Exhibit 14B. As support to this topic, Cell D10 of the Adjusted Penetration Row (see **Exhibit TN6** [Case Exhibit 14B completed]) has been added as a modifiable field that updates the spreadsheet results based on changing the market penetration assumption (100 percent indicates the full original assumption; 50 percent indicates half of the original assumption). After modifying this cell to read 50 percent, switch to the solution spreadsheet of Case Exhibit 14A to display how the project's ROI is impacted by the resulting market penetration change.

The key parameter is market penetration. If the original assumed market penetration is decreased from 100 percent to 50 percent, the ROI falls from 40.6 percent to 9.2 percent. In addition, this assumes a five-year time period for the analysis. Varying this single parameter acts as a basic introduction to sensitivity analysis.

As noted previously, a three-year time frame is more appropriate than five years, since for a Web portal initiative the infrastructure will most likely be replaced or upgraded within three years. In general, managers should choose consistent time frames for ROI analysis in order to objectively compare projects using IRR. The five-year ROI for full assumptions is calculated at the bottom of the solution spreadsheet in Case Exhibit 14A (see Exhibit TN1 for Case Exhibit 14A completed). At the bottom of the spreadsheet (Rows 33 to 36) are IRR and payback figures based on a three-year time frame (see **Exhibit TN7**). Clearly the three-year IRR is less than the hurdle rate for the firm. Furthermore, if the market penetration assumption is reduced to 50 percent, the IRR is actually so bad it will not calculate.

After the students understand the importance of basic sensitivity analysis and the time frame of the IRR calculation, the instructor can walk through the more sophisticated multi-dimensional sensitivity tables. The five-year single-variable sensitivity and five-year multiple-variable sensitivity spreadsheets in Exhibits TN2, TN3, TN4, and TN5 illustrate how different changes to the case assumptions impact the IRR and NPV of the project. The sensitivity analysis is very easy

to accomplish using the Table function in Excel.² The cells are color-coded so that green represents an IRR greater than the firm hurdle rate and red less than the hurdle rate.

The single-variability sensitivity clearly shows the IRR is not impacted if the assumptions for tier migration, order size, and order channel are not met. The cells are green with the IRR greater than the firm hurdle rate for almost all possible values of the assumptions. However, the market penetration assumption has significant impact on the IRR. This is noted by the 50 percent red cells in the exhibit. Clearly, market penetration is the key assumption in this ROI model. The two-dimensional sensitivity gives insight into the tradeoff between assumptions in the model.

Once the students understand the concept of sensitivity analysis, come back to the time frame issue by referring the class to the additional computations included in the solution spreadsheet of Case Exhibit 14A. After the students understand how this was calculated, present the three-year single-variable sensitivity and three-year multiple-variable sensitivity spreadsheets to illustrate how changes in the time frame impact the IRR and NPV of the project. Note that all the cells are red with a three-year time frame. Hence, the three-year ROI is not acceptable.

At this point, the instructor may ask the class if they would still continue with the project given the new findings presented. Most, if not all of them, will probably decide not to continue with the initiative. Open a new discussion by asking them what qualitative or quantitative steps they would take to improve the financials and increase the success potential of the project. To conclude the discussion, relay to the class Anfield's expert comments regarding the case (see the following section). Finally, summary slides are provided for a discussion of the key takeaways of the case.

Executive Expert Comments

A digital video accompanies this case showing Anfield debriefing the case and providing his expert comments regarding the project; this video can be shown to the students. The companies involved in the case, B&K Distributors and JDA Consulting, are real, but the names have been changed for confidentiality. Anfield was the director of business development for the consulting company hired by B&K Distributors for this initiative, and he had firsthand experience on this project.

First, Anfield clarified that the assumptions in the case were actually provided by B&K Distributors's senior management team. He then explained that as the consultant for the project, his role was to question and analyze the figures for feasibility. Even though these numbers were dictated by senior management, the team still needed to conduct a reality check to determine if they were achievable.

Second, Anfield explained that as a consultant he would never have provided senior management with the initial ROI figures the template exhibits calculate (the results of Step 1). By conducting the appropriate sensitivity analysis with a three-year time frame, he realized that the success of the project depended largely on the assumptions made by senior management as well as the implementation and annual maintenance costs. He focused his attention on the component that he could control—the costs. He decided to re-plan the project with lower cost figures by

² The official Excel documentation for the Table function is a little confusing. Hence, a short tutorial for students is provided as a handout in the Appendix. This handout can be passed out after the class so that students can use the technique to analyze future cases.

reducing the scope of the project. This allowed the project to have a more feasible ROI that senior management was very comfortable with. For Anfield, this was also a strategic move for his consulting firm. By providing the discount and showing good faith to B&K Distributors, he was able to develop a long-term relationship with the company and secure future partnership deals.

To illustrate, reductions in the costs of the project dramatically improve the ROI. For example, reducing the cost of the project in the accompanying spreadsheet solution file can be achieved by cutting \$100,000 from the marketing budget and reducing the scope of the project to lower the implementation costs, which results in additional project cost savings of \$100,000. With this adjustment, the IRR becomes greater than the hurdle rate for even the three-year analysis.

Finally, Anfield explained that the initiative did not push through. Although the figures and assumptions presented to senior management were conservative and feasible, B&K Distributors still decided not to continue with the project due to several internal corporate reasons. It turned out, though, that had they continued forward with the project, the actual order channel migration rate of customers from regular channels to Web-based portals would have been higher than the team originally expected.

Summary and Key Takeaways

Provided with this teaching note is a PowerPoint presentation that summarizes the case problems that need to be addressed, the main points of the ROI analysis, and the key takeaways of the case (**Exhibit TN8**). These slides may be used to reiterate the advantages of applying a comprehensive sensitivity analysis to an ROI activity and experimenting with different project time frames that are appropriate to the case at hand.

As mentioned at the start of this teaching note, the key learning of this case is that ROI analysis is not a straightforward activity—several factors need to be analyzed to be able to conduct a thorough review of the feasibility of the investment.

First, it is always better to use conservative assumptions when calculating a project's ROI. At the same time, it is also essential for the team to discuss and finalize these assumptions with senior management before the start of the project to make sure that expectations and forecasts are well managed. If the assumptions are created in partnership with the senior management, then the ROI analysis will ultimately be more believable. Remember, it is better to under-promise initially and over-deliver later on than to over-promise initially and then fall short or fail to meet senior management expectations later on.

Second, if any ROI calculation sensitivity analysis is essential, see the Appendix for instructions. By conducting this type of an analysis, the team is forced to identify the major assumptions of the model and recognize the risks that may impact the results of the project. Once the team understands all the potential scenarios that may occur, they can isolate the critical parameters of the ROI model and focus on managing these issues to improve the project's probability of success. In addition, the analysis can also help the team experiment on the flexibility of the projections given different scenarios. Finally, the sensitivity analysis may also provide clues on how the project business case should be changed to improve the final ROI of the project.

Third, it is always better to estimate or identify ranges when calculating a project's ROI. Since the team is making a forecast that will occur at a later time, it is impossible to predict the financial outcome of the project perfectly. By identifying ranges, the team is making an educated guess based on the best-case, expected-case, and worst-case scenarios that may happen. This gives the team more flexibility in terms of managing the expectations of the company's executives. However, it is also important for the team to consider which results to present to senior management for final approval of the project.

Fourth, the length of time used in the financial analysis of a project significantly impacts the final ROI. Utilizing different time frames (for example, three years versus five years) may mean the difference between approving and rejecting a project, so it is important to choose a period of time that makes sense. This is extremely important for technology-based projects, since technology improvements occur very rapidly, and ROI decisions may have to be based on shorter time frames as compared to other, more constant capital investment projects.

Finally, it is also important to consider the qualitative issues of a project. For example, how essential is the project in terms of the company's overall strategy? Does the project have any additional option value that may impact its evaluation? The ROI figures assist the team in quantifying the purpose of the project and understanding the financial risks and parameters of various scenarios that may occur. However, the final recommendation will have to be made based on how these figures are analyzed and transformed into a comprehensive business case that makes sense to the company's executives in the context of the strategic and business objectives of the firm.



Exhibit TN1: Five-Year Internal Rate of Return**B&K Distributors—ROI Model—Summary Page (US\$ in thousands)**

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue						
Upside		4,643	5,619	6,487	7,293	7,886
Baseline		3,608	3,828	4,061	4,308	4,569
Incremental		1,035	1,791	2,426	2,986	3,316
Net income						
Upside		583	734	878	1,003	1,121
Baseline		436	463	491	521	552
Incremental		147	272	387	482	569
Incremental investment program (after tax)						
Upfront costs	(387)					
Ongoing costs		(283)	(291)	(300)	(309)	(319)
Total	(387)	(283)	(291)	(300)	(309)	(319)
Total (after tax)	(240)	(175)	(181)	(186)	(192)	(197)
Total incremental cash flow	(240)	(28)	91	201	291	371
		(268)	(177)	24	314	686
Five-year internal rate of return	40.6%					
Net present value	346					
Discount rate	12%					
Tax rate	38%					
Support cost inflation rate	3%					
THREE-YEAR IRR	3.3%					
Total incremental cash flow	(240)	(28)	91	201	291	371
Cumulative cash flows		(268)	(177)	24	314	686
Payback				0.88		

Exhibit TN2: Single-Variable Sensitivity (Five Years)**(As a Percentage of Upside Projections)**

PENETRATION		TIER MIGRATION		ORDER SIZE		ORDER CHANNEL	
100%	40.7%	100%	40.7%	100%	40.7%	100%	40.7%
90%	34.9%	90%	38.6%	90%	38.1%	90%	39.4%
80%	29.0%	80%	36.6%	80%	35.4%	80%	38.2%
70%	22.7%	70%	34.5%	70%	32.7%	70%	36.9%
60%	16.2%	60%	32.4%	60%	29.9%	60%	35.6%
50%	9.2%	50%	30.2%	50%	27.0%	50%	34.4%
40%	1.6%	40%	28.0%	40%	24.0%	40%	33.1%
30%	-6.8%	30%	25.8%	30%	20.9%	30%	31.7%
20%	-16.3%	20%	23.5%	20%	17.6%	20%	30.4%
10%	#NUM!	10%	21.1%	10%	14.2%	10%	29.0%
0%	#NUM!	0%	18.7%	0%	10.6%	0%	27.6%

Red; indicates results below the hurdle rate.

Green; indicates results above the hurdle rate.

Exhibit TN3: Multiple-Variable Sensitivity (Five Years)**(As a Percentage of Upside Projections)**

PENETRATION								
TIER MIGRATION		100%	90%	80%	70%	60%	50%	40%
	100%	40.7%	34.9%	29.0%	22.7%	16.2%	9.2%	1.6%
	80%	36.6%	30.8%	24.7%	18.4%	11.7%	4.4%	-3.5%
	60%	32.4%	26.5%	20.3%	13.9%	6.9%	-0.6%	-8.9%
	50%	30.2%	24.3%	18.1%	11.5%	4.4%	-3.2%	-11.8%
	40%	28.0%	22.0%	15.8%	9.1%	1.9%	-6.0%	-14.8%
	20%	23.5%	17.4%	10.9%	4.0%	-3.5%	-11.9%	#NUM!
	0%	18.7%	12.5%	5.8%	-1.4%	-9.4%	-18.4%	#NUM!

PENETRATION								
ORDER SIZE		100%	90%	80%	70%	60%	50%	40%
	100%	40.7%	34.9%	29.0%	22.7%	16.2%	9.2%	1.6%
	80%	35.4%	29.5%	23.4%	16.9%	10.0%	2.5%	-5.7%
	60%	29.9%	23.8%	17.4%	10.6%	3.2%	-4.9%	-14.0%
	50%	27.0%	20.8%	14.2%	7.2%	-0.5%	-9.0%	-18.9%
	40%	24.0%	17.6%	10.9%	3.6%	-4.4%	-13.5%	#NUM!
	20%	17.6%	10.9%	3.6%	-4.4%	-13.4%	#NUM!	#NUM!
	0%	10.6%	3.4%	-4.7%	-13.8%	#NUM!	#NUM!	#NUM!

PENETRATION								
ORDER CHANNEL		100%	90%	80%	70%	60%	50%	40%
	100%	40.7%	34.9%	29.0%	22.7%	16.2%	9.2%	1.6%
	80%	38.2%	32.4%	26.4%	20.1%	13.4%	6.2%	-1.6%
	60%	35.6%	29.8%	23.7%	17.3%	10.5%	3.2%	-4.9%
	50%	34.4%	28.5%	22.4%	15.9%	9.0%	1.6%	-6.6%
	40%	33.1%	27.1%	21.0%	14.4%	7.5%	-0.1%	-8.4%
	20%	30.4%	24.4%	18.1%	11.5%	4.3%	-3.5%	-12.2%
	0%	27.6%	21.6%	15.2%	8.4%	1.0%	-7.1%	-16.3%

 Red; indicates results below the hurdle rate.

 Green; indicates results above the hurdle rate.

Exhibit TN4: Single-Variable Sensitivity (Three Years)**(As a Percentage of Upside Projections)**

PENETRATION		TIER MIGRATION		ORDER SIZE		ORDER CHANNEL	
100%	3.3%	100%	3.3%	100%	3.3%	100%	3.3%
90%	-3.8%	90%	0.9%	90%	0.6%	90%	1.9%
80%	-11.3%	80%	-1.6%	80%	-2.2%	80%	0.4%
70%	-19.3%	70%	-4.2%	70%	-5.0%	70%	-1.1%
60%	-27.9%	60%	-6.8%	60%	-7.9%	60%	-2.7%
50%	#NUM!	50%	-9.5%	50%	-10.9%	50%	-4.3%
40%	#NUM!	40%	-12.2%	40%	-14.1%	40%	-5.9%
30%	#NUM!	30%	-15.1%	30%	-17.3%	30%	-7.5%
20%	#NUM!	20%	-18.0%	20%	-20.6%	20%	-9.1%
10%	#NUM!	10%	-21.0%	10%	-24.1%	10%	-10.8%
0%	#NUM!	0%	-24.1%	0%	-27.7%	0%	-12.5%

 Red; indicates results below the hurdle rate.

 Green; indicates results above the hurdle rate.

Exhibit TN5: Multiple-Variable Sensitivity (Three Years)**(As a Percentage of Upside Projections)**

		PENETRATION						
		100%	90%	80%	70%	60%	50%	40%
TIER MIGRATION	100%	3.3%	−3.8%	−11.3%	−19.3%	−27.9%	#NUM!	#NUM!
	80%	−1.6%	−8.9%	−16.6%	−24.9%	#NUM!	#NUM!	#NUM!
	60%	−6.8%	−14.3%	−22.3%	#NUM!	#NUM!	#NUM!	#NUM!
	50%	−9.5%	−17.1%	−25.2%	#NUM!	#NUM!	#NUM!	#NUM!
	40%	−12.2%	−20.0%	−28.3%	#NUM!	#NUM!	#NUM!	#NUM!
	20%	−18.0%	−26.0%	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!
	0%	−24.1%	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!
		PENETRATION						
		100%	90%	80%	70%	60%	50%	40%
ORDER SIZE	100%	3.3%	−3.8%	−11.3%	−19.3%	−27.9%	#NUM!	#NUM!
	80%	−2.2%	−9.5%	−17.3%	−25.7%	#NUM!	#NUM!	#NUM!
	60%	−7.9%	−15.5%	−23.7%	#NUM!	#NUM!	#NUM!	#NUM!
	50%	−10.9%	−18.7%	−27.1%	#NUM!	#NUM!	#NUM!	#NUM!
	40%	−14.1%	−22.0%	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!
	20%	−20.6%	−29.0%	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!
	0%	−27.7%	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!	#NUM!
		PENETRATION						
		100%	90%	80%	70%	60%	50%	40%
ORDER CHANNEL	100%	3.3%	−3.8%	−11.3%	−19.3%	−27.9%	#NUM!	#NUM!
	80%	0.4%	−6.9%	−14.5%	−22.7%	#NUM!	#NUM!	#NUM!
	60%	−2.7%	−10.0%	−17.9%	−26.3%	#NUM!	#NUM!	#NUM!
	50%	−4.3%	−11.7%	−19.6%	−28.1%	#NUM!	#NUM!	#NUM!
	40%	−5.9%	−13.3%	−21.4%	#NUM!	#NUM!	#NUM!	#NUM!
	20%	−9.1%	−16.8%	−25.0%	#NUM!	#NUM!	#NUM!	#NUM!
	0%	−12.5%	−20.4%	−28.9%	#NUM!	#NUM!	#NUM!	#NUM!

Red; indicates results below the hurdle rate.

Green; indicates results above the hurdle rate.

Exhibit TN6: Sensitivity Analysis to Adjusted Penetration**B&K Distributors—ROI Model—Upside Projection**

(Numbers in thousands)		Year 2	Year 3	Year 4	Year 5
Number of franchise locations		3,608	3,608	3,680	3,754
Growth		2.0%	2.0%	2.0%	2.0%
Penetration		60.0%	65.0%	68.0%	70.0%
Adjusted penetration	100%	50.0%	60.0%	65.0%	68.0%
Number of customers		1,700	2,081	2,299	2,454
Tier structure					
Percent		25%	28%	30%	32%
A # of locations		425	583	690	785
Order frequency		12	12	12	12
Percent		55%	58%	60%	60%
B # of locations		935	1,207	1,380	1,472
Order frequency		8	8	8	8
Percent		20%	14%	10%	8%
C # of locations		340	291	230	196
Order frequency		3	3	3	3
Average order size		\$250.00	\$265.00	\$280.90	\$297.75
Growth			6.0%	6.0%	6.0%
Total transactions					
A accounts		5,100	6,991	8,277	9,421
B accounts		7,480	9,655	11,037	11,777
C accounts		1,020	874	690	589
Total		13,600	17,520	20,004	21,787
Revenue					
A accounts		1,275	1,853	2,325	2,805
B accounts		1,870	2,559	3,100	3,507
C accounts		255	232	194	175
Total		3,400	4,643	5,619	6,487
Unit margin					
COGS %		68.70%	68.70%	68.70%	68.70%
Gross margin per order		78.25	82.95	87.92	93.20
Processing cost per order		29.50	29.25	28.71	28.19
Net contribution per order		48.75	53.70	59.21	65.01
Account contribution					
Transactions		13,600	17,520	20,004	21,787
Net margin per order		49	54	59	65
Operating income		663	941	1,184	1,416
Tax					
Tax rate		38%	38%	38%	38%
Tax		252	357	450	538
Net contribution after tax		411	583	734	878

Change this parameter to reduce the market penetration assumption

Exhibit TN7: Three-Year Internal Rate of Return

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
THREE-YEAR IRR	3.3%					
Total incremental cash flow	(240)	(28)	91	201	291	371
Cumulative cash flows		(268)	(177)	24	314	686
		<i>Payback</i> (near end of Year 3)		0.88		

Exhibit TN8: Summary Slides

B&K Distributors: ROI for a Web-Based Customer Portal

- Designed to be a self-contained introduction to ROI analysis
- Technology is for a Web portal—easy to understand
- Medium-sized business with \$100 million in revenues
- Consultant trying to “sell” consulting services
- ROI templates given so that the basic analysis follows a recipe format
- But significant learnings in the case . . .

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Exhibit TN8 (continued)

B&K Distributors Case Five-Year ROI

Single-Variable Sensitivity (as a Percentage of Upside Projections)							
PENETRATION		TIER MIGRATION		ORDER SIZE		ORDER CHANNEL	
100%	40.7%	100%	40.7%	100%	40.7%	100%	40.7%
90%	34.9%	90%	38.6%	90%	38.1%	90%	39.4%
80%	29.0%	80%	36.6%	80%	35.4%	80%	38.2%
70%	22.7%	70%	34.5%	70%	32.7%	70%	36.9%
60%	16.2%	60%	32.4%	60%	29.9%	60%	35.6%
50%	9.2%	50%	30.2%	50%	27.0%	50%	34.4%
40%	1.6%	40%	28.0%	40%	24.0%	40%	33.1%
30%	-6.8%	30%	25.8%	30%	20.9%	30%	31.7%
20%	-16.3%	20%	23.5%	20%	17.6%	20%	30.4%
10%	#NUM!	10%	21.1%	10%	14.2%	10%	29.0%
0%	#NUM!	0%	18.7%	0%	10.6%	0%	27.6%

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Three-Year ROI

Single-Variable Sensitivity (as a Percentage of Upside Projections)							
PENETRATION		TIER MIGRATION		ORDER SIZE		ORDER CHANNEL	
100%	3.3%	100%	3.3%	100%	3.3%	100%	3.3%
90%	-3.8%	90%	0.9%	90%	0.6%	90%	1.9%
80%	-11.3%	80%	-1.6%	80%	-2.2%	80%	0.4%
70%	-19.3%	70%	-4.2%	70%	-5.0%	70%	-1.1%
60%	-27.9%	60%	-6.8%	60%	-7.9%	60%	-2.7%
50%	#NUM!	50%	-9.5%	50%	-10.9%	50%	-4.3%
40%	#NUM!	40%	-12.2%	40%	-14.1%	40%	-5.9%
30%	#NUM!	30%	-15.1%	30%	-17.3%	30%	-7.5%
20%	#NUM!	20%	-18.0%	20%	-20.6%	20%	-9.1%
10%	#NUM!	10%	-21.0%	10%	-24.1%	10%	-10.8%
0%	#NUM!	0%	-24.1%	0%	-27.7%	0%	-12.5%

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Exhibit TN8 (continued)

Three-Year ROI by Reducing Scope and Implementation Costs

Multiple-Variable Sensitivity (as a Percentage of Upside Projections)								
TIER MIGRATION	PENETRATION							
	100%	90%	80%	70%	60%	50%	40%	
	100%	62.1%	54.4%	46.6%	38.5%	30.2%	21.6%	12.6%
	80%	56.9%	49.2%	41.3%	33.2%	24.8%	16.1%	6.9%
	60%	51.6%	43.9%	36.0%	27.8%	19.3%	10.4%	1.0%
	50%	48.9%	41.2%	33.2%	25.0%	16.4%	7.5%	-2.0%
	40%	46.1%	38.4%	30.4%	22.1%	13.5%	4.5%	-5.2%
	20%	40.6%	32.8%	24.7%	16.3%	7.5%	-1.7%	-11.7%
	0%	34.9%	27.0%	18.8%	10.3%	1.3%	-8.3%	-18.6%
ORDER SIZE	PENETRATION							
	100%	90%	80%	70%	60%	50%	40%	
	100%	62.1%	54.4%	46.6%	38.5%	30.2%	21.6%	12.6%
	80%	56.8%	49.1%	41.2%	33.1%	24.7%	15.9%	6.7%
	60%	51.5%	43.7%	35.8%	27.5%	19.0%	10.0%	0.5%
	50%	48.7%	41.0%	33.0%	24.7%	16.0%	6.9%	-2.7%
	40%	46.0%	38.2%	30.1%	21.7%	13.0%	3.8%	-6.0%
	20%	40.4%	32.5%	24.3%	15.8%	6.8%	-2.7%	-12.9%
	0%	34.6%	26.6%	18.2%	9.5%	0.3%	-9.6%	-20.3%

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Summary ROI Analysis Framework

1. Business Discovery: Understand the business, and the cost and revenue drivers that the new technology project is expected to impact. Make judgments and/or assumptions about how the technology project will impact cost and revenue drivers to improve business performance.

2. Base Case: Determine base case cash flows as if the firm will continue its operations without implementing the technology project.

3. Project Costs: Determine costs associated with implementing the technology project. These costs involve both the initial investment and recurring maintenance costs.

4. Free Cash Flows with Project: Determine free cash flows after the firm has implemented the technology project, based on assumptions for the business drivers and costs of the project.

5. Incremental Cash Flows: Determine the incremental cash flows by subtracting the base cash flows from the cash flows in Step 4. Calculate NPV, IRR, and payback period.

6. Sensitivity Analysis: Perform sensitivity analysis (Excel tools/Monte Carlo simulations) to incorporate varying assumptions and risk factors to understand the range of possible outcomes.

Traditional
ROI/NPV Analysis

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See [ROI Analysis](#) chapter by Jeffery in the Wiley Encyclopedia of the Internet for the detailed discussion

Exhibit TN8 *(continued)*

B&K Case Takeaways

- Use conservative assumptions derived in partnership with management
- The length of time used for the analysis significantly impacts the ROI
 - Choose a period of time that makes sense
 - Calculate 1, 2, 3 & 5 years where appropriate
- For any ROI calculation, sensitivity analysis is essential
- Find the critical parameters in the model
- Understand risks
- Sensitivity analysis gives clues on how the plan should be changed to improve the ROI
- Always calculate the range of ROIs
 - Worst, best, and expected case
 - Consider which results you will present to management
- If the ROI is less than the hurdle rate, ask “How essential is this project for our corporate strategy? Can we reduce scope to deliver the majority of benefits?”
- Ask “Does the project have additional option value that may factor into the decision?”

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Appendix: How to Perform Sensitivity Analysis in Excel with a Data Table

Synopsis

This appendix answers a common student question: How to create a multi-dimensional data table in Excel for a return on investment sensitivity analysis? This note accompanies the B&K Distributors Case, and is handed out as part of the case debrief.

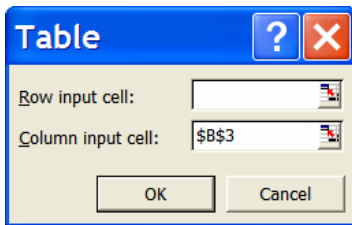
One-Variable Data Table

The input cell used in our example is cell B3, also referred to as “Variable A.” In our example, the input cell is a factor that is multiplied by each of cost of goods sold (COGS) values in cells B6:E6. The net profit is the difference between the sales revenue and the COGS. The net present value (NPV) in cell B9 is a traditional NPV calculated using the following formula: =NPV(0.1,B7:E7). One-variable data table values are listed either down a column (column-oriented) or across a row (row-oriented). Formulas used in a one-variable data table must refer to an input cell for your spreadsheet model.

	A	B	C	D	E
1	Example 1: A 1-Dimensional Table				
2					
3	Variable A	1			
4					
5	Sales Revenue	10	14	16	25
6	COGS	7	12	13	22
7	Net Profit	3	2	3	3
8					
9	NPV	\$8.7			
10					

1. Type the list of values you want to substitute in the input cell either down one column or across one row. Our example will use column orientation, and our input values will begin with 1.0 and increase by 0.1 until we reach 1.5. Type these values in cells A13:A18. These values will be substituted into the input cell, B3 (Variable A) by the data table function.
2. Type the formula for which you wish to determine the sensitivity in the row above the first value and one cell to the right of the column of values. In our example, we want to determine how the NPV changes when Variable A changes. Type =B9 into cell B12 for our example.
3. Select the range of cells that contain the formulas and values you want to substitute. For our example, highlight cells A12:B18.
4. From the menus choose *Data, Table*. Excel displays the “Table” dialog that asks you to identify the cell reference for the input cell. If the data table is column-oriented, type the cell reference for the input cell in the “Column input cell” box. For the example below, type \$B\$3 in the “Column input cell” box and leave the “Row input cell” box blank. If the data table is

row-oriented, type the cell reference for the input cell in the “Row input cell” box and leave the “Column input cell” box blank.



5. Click “OK” to have Excel execute the data table and fill the data table with values. The output should look like the diagram below.

	A	B
12		\$8.7
13	1.0	\$8.7
14	1.1	\$4.6
15	1.2	\$0.5
16	1.3	-\$3.6
17	1.4	-\$7.7
18	1.5	-\$11.9

6. To color code the output, highlight the output cells, B13:B18. From the menus choose *Format, Conditional Formatting*. Excel displays the “Conditional Formatting” dialog that asks you for conditions. In the second box from the left, change “between” to “less than” by using the drop-down arrow. In the cell immediately to the right, enter the value “0.” Finally, click on the “Format . . .” button to reveal the “Format Cells” dialog. Click on the “Patterns” tab, and select the color red and then click “OK.”
7. Now, click on the “Add >>” button on the “Conditional Formatting” dialog to add another condition. A second condition is added to the dialog. Again, in the Condition 2 section, modify the second box from the left. Change “between” to “greater than or equal to” by using the drop-down arrow. In the cell immediately to the right, enter the value “0.” Finally, click on the “Format . . .” button to reveal the “Format Cells” dialog. Click on the “Patterns” tab, select the color green, and then click “OK.” Finally, on the “Conditional Formatting” dialog press the “OK” button.

Conditional Formatting

Condition 1
 Cell Value Is less than 0
 Preview of format to use when condition is true: AaBbCcYyZz (Red background)
 Format...

Condition 2
 Cell Value Is greater than or equal to 0
 Preview of format to use when condition is true: AaBbCcYyZz (Green background)
 Format...

Buttons: Add >>, Delete..., OK, Cancel

8. Your final output should appear as below.

	A	B	C	D	E
1	Example 1: A 1-Dimensional Table				
2					
3	Variable A	1			
4					
5	Sales Revenue	10	14	16	25
6	COGS	7	12	13	22
7	Net Profit	3	2	3	3
8					
9	NPV	\$8.7			
10					
11					
12		\$8.7			
13	1.0	\$8.7			
14	1.1	\$4.6			
15	1.2	\$0.5			
16	1.3	-\$3.6			
17	1.4	-\$7.7			
18	1.5	-\$11.9			

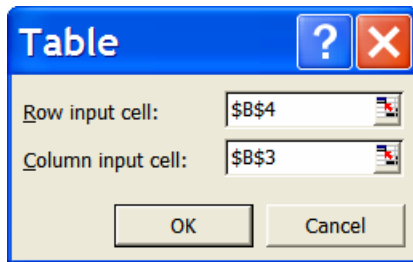
The function you want to calculate goes here.

Two-Variable Data Table

For our two-variable example we will need two input variables. We will again use “Variable A,” cell B3, as our first input cell. The second input cell we will use is cell B4, also referred to as “Variable B.” As before, the Variable A input cell is a factor that is multiplied by each of the COGS values in cells B7:E7. The Variable B input cell is a factor that is multiplied by each of the sales revenue values in cells B6:E6. The net profit is the difference between the sales revenue and the COGS. The NPV in cell B10 is a traditional NPV calculated using the following formula: =NPV(0.1,B8:E8). Two-variable data table values are listed in a column (column-oriented) and across a row (row-oriented). Formulas used in a two-variable data table must refer to two inputs cell for your spreadsheet model.

	A	B	C	D	E
1	Example 2: A 2-Dimensional Table				
2					
3	Variable A	1			
4	Variable B	1			
5					
6	Sales Revenue	10	14	16	25
7	COGS	7	12	13	22
8	Net Profit	3	2	3	3
9					
10	NPV	\$8.7			
11					

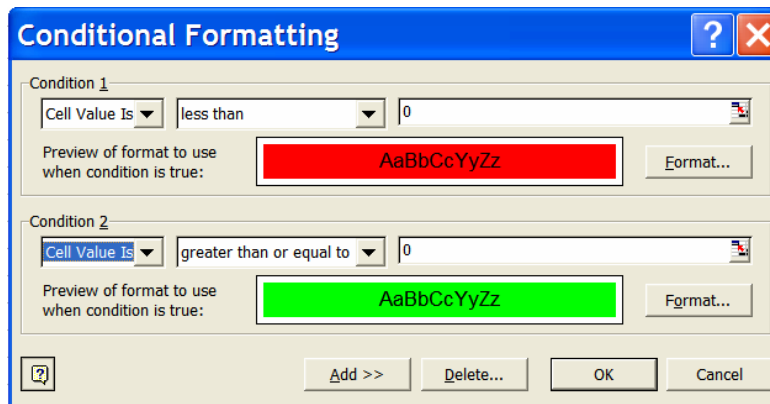
1. Type the list of values you want to substitute in the first input cell down a column and the second input cell across a row. Our example will use input values that begin with 1.0 and increase by 0.1 until we reach 1.5 for both inputs. Type these values in cells A16:A21; these values will be substituted into the first input cell, B3 (Variable A). Again, type these values in cells B15:G15; these values will be substituted into the second input cell, B4 (Variable B).
2. In a two-variable data table, type the formula in the “intersection cell” of the column values and row values. For our example we want to determine how the NPV changes when Variable A and Variable B change. Type =B10 into cell A15 for our example.
3. Select the range of cells that contains the formulas and values you want to substitute. For our example, highlight cells A15:G21.
4. From the menus choose *Data, Table*. Excel displays the “Table” dialog that asks you to identify the cell reference for the input cell. For a two-variable data table, type the cell reference for the first input cell (Variable A) in the “Column input cell” box. For the example below, type \$B\$3 in the “Column input cell” box and type \$B\$4 in the “Row input cell.”



5. Click “OK” to have Excel execute the data table and fill the data table with values.

	A	B	C	D	E	F	G
14				Variable B			
15	\$8.7	1	1.1	1.2	1.3	1.4	1.5
16	Variable A	1.0	8.7	13.7	18.6	23.6	33.6
17		1.1	4.6	9.6	14.5	19.5	29.5
18		1.2	0.5	5.4	10.4	15.4	25.3
19		1.3	-3.6	1.3	6.3	11.3	21.2
20		1.4	-7.7	-2.8	2.2	7.2	12.2
21		1.5	-11.9	-6.9	-1.9	3.1	8.0

6. To color code the output, highlight the output cells, which are cells B16:G21 in the example below. From the menus choose *Format, Conditional Formatting*. Excel displays the “Conditional Formatting” dialog that asks you for conditions. In the second box from the left, change “between” to “less than” by using the drop-down arrow. In the cell immediately to the right, enter the value “0.” Finally, click on the “Format . . .” button to reveal the “Format Cells” dialog. Click on the “Patterns” tab, select the color red, and then click “OK.”
7. Now, click on the “Add >>” button on the “Conditional Formatting” dialog to add another condition. A second condition is added to the dialog. Again, in the “Condition 2” section, modify the second box from the left. Change “between” to “greater than or equal to” by using the drop-down arrow. In the cell immediately to the right, enter the value “0.” Finally, click on the “Format . . .” button to reveal the “Format Cells” dialog. Click on the “Patterns” tab, select the color green, and then click “OK.” Finally, on the “Conditional Formatting” dialog, press the “OK” button.



8. Your final output should appear as below.

	A	B	C	D	E	F	G
1	Example 2: A 2-Dimensional Table						
2							
3	Variable A	1					
4	Variable B	1					
5							
6	Sales Revenue	10	14	16	25		
7	COGS	7	12	13	22		
8	Net Profit	3	2	3	3		
9							
10	NPV	\$8.7					
11							
12							
13							
14				Variable B			
15	\$8.7	1	1.1	1.2	1.3	1.4	1.5
16	Variable A	1.0	8.7	13.7	18.6	23.6	33.6
17		1.1	4.6	9.6	14.5	19.5	24.5
18		1.2	0.5	5.4	10.4	15.4	20.4
19		1.3	-3.6	1.3	6.3	11.3	16.3
20		1.4	-7.7	-2.8	2.2	7.2	12.2
21		1.5	-11.9	-6.9	-1.9	3.1	8.0

The function you want to calculate goes here.